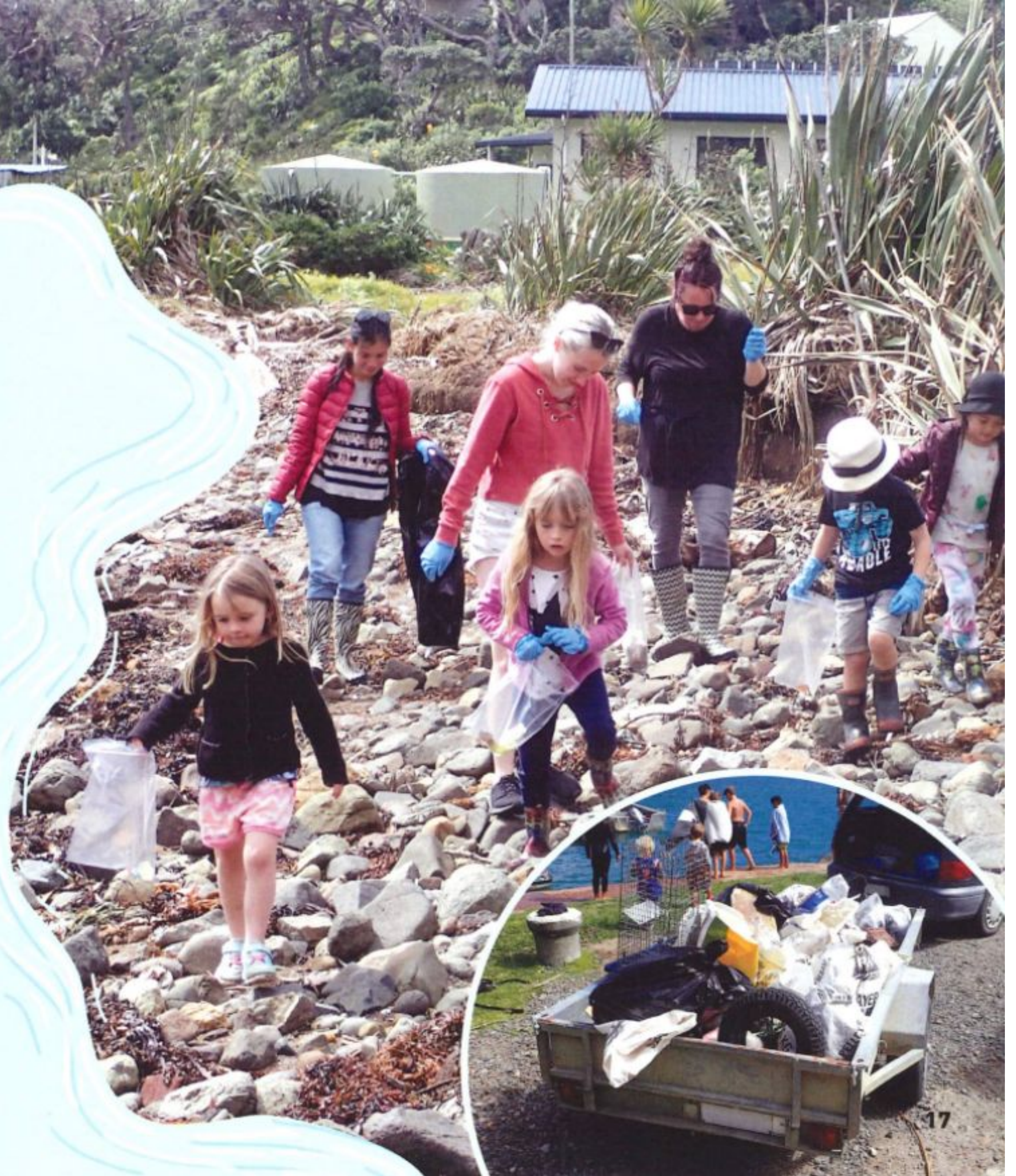
SEA SCIENCE

by Dr Marie McEntee, Joe Fagan, and the schools of Aotea

Rubbish. It's easy to make but hard to get rid of. For too long, students on Aotea Great Barrier Island have watched rubbish wash up on their beautiful beaches. In 2017, they all became scientists and did something about it.



WHAT IS MARINE DEBRIS?

Rubbish that enters the ocean is called **marine** debris. It includes things like food wrappers, bottle caps, straws, and fishing gear – and 60 to 95 percent of it is plastic. Plastic is especially damaging because it never disappears. Instead, it breaks down into smaller and smaller pieces. These small pieces create lots of big problems.

Why don't we know for sure how much of all marine debris is plastic? The ocean is a very large place. It would be impossible to search through all of it. Instead, scientists count the rubbish in small sample areas. In some places, over half of the marine debris they find is plastic. In others, it's almost all plastic. From this information, scientists can predict how much plastic is in the rest of the ocean.

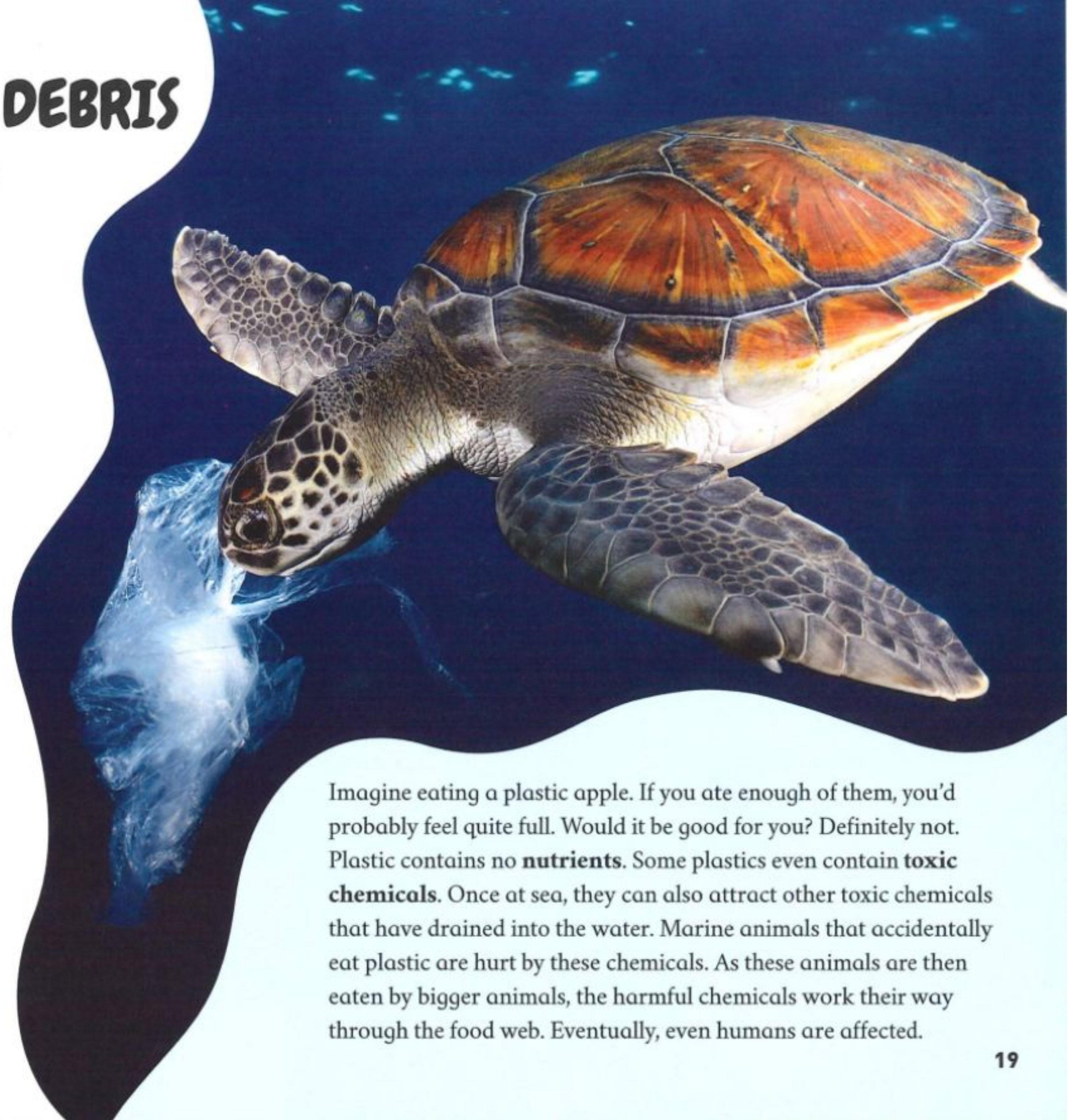
The Great Pacific Garbage Patch

Scientists estimate that at least 8 million tonnes of plastic enter our oceans every year. That's the same as one rubbish truck dumping a full load of plastic into the ocean every minute. Where does it all go? A lot of this rubbish is picked up by large circular ocean currents called gyres. The North Pacific gyre in the Pacific Ocean has gathered trillions of pieces of rubbish into a huge area called the Great Pacific Garbage Patch. It's the largest collection of ocean plastic in the world. You might imagine the Great Pacific Garbage Patch to look like a floating island of bottles and wrappers, but actually, it looks more like plastic soup.

THE DANGERS OF DEBRIS

Marine debris harms our oceans and marine life in lots of different ways.

- Birds, fish, and marine mammals become trapped in marine debris and mistake it for food.
- Marine debris collects at the ocean's surface and blocks out sunlight. This affects **organisms** that need light to survive. As soon as one organism is affected, the other animals who feed on that organism are also affected.
- Shellfish and other animals can attach themselves to floating pieces of plastic. Sometimes, ocean currents can carry these "plastic hitchhikers" into a new **environment**. This can cause problems for the organisms who already live there.
- Marine debris also damages boats and pollutes beaches.

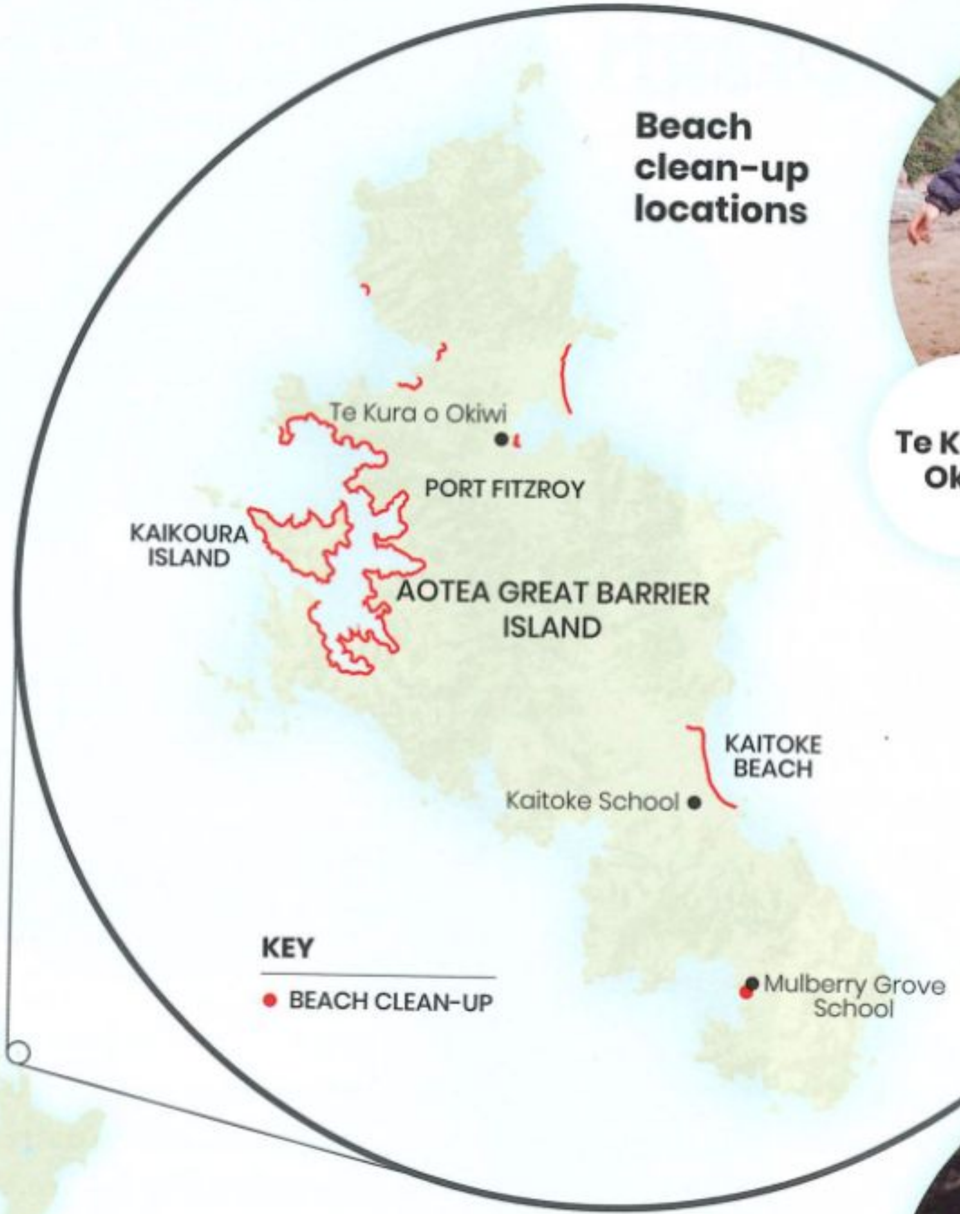


Imagine eating a plastic apple. If you ate enough of them, you'd probably feel quite full. Would it be good for you? Definitely not. Plastic contains no **nutrients**. Some plastics even contain **toxic chemicals**. Once at sea, they can also attract other toxic chemicals that have drained into the water. Marine animals that accidentally eat plastic are hurt by these chemicals. As these animals are then eaten by bigger animals, the harmful chemicals work their way through the food web. Eventually, even humans are affected.

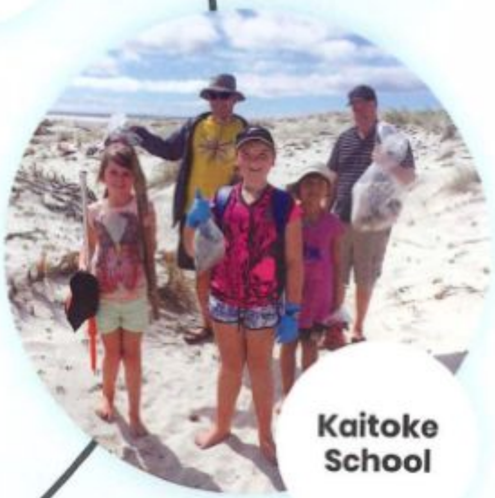
ASKING QUESTIONS

In 2017, students from Te Kura o Okiwi, Mulberry Grove School, and Kaitoke School decided to investigate the marine debris washing up on Aotea Great Barrier Island. Working with scientists from the University of Auckland, they came up with some big questions that they wanted to answer:

- What kinds of marine debris do we find on our beaches?
- Where does it come from?
- Does the location of the beach affect the type of marine debris found?
- How can we reduce the amount of marine debris on our island?




Te Kura o Okiwi


Kaitoke School


Mulberry Grove School

GATHERING DATA

The schools decided that the best way to answer these questions was to do a beach clean-up. They split up and cleaned the beaches closest to their schools. Te Kura o Okiwi cleaned the bays and harbour beaches in the north. Kaitoke School cleaned Kaitoke Beach, a long, sandy surf beach facing the Pacific Ocean. Mulberry Grove School cleaned a sheltered bay in the south.

The students were given compasses to record the direction their beach was facing. A compass is a tool containing a needle that always points north. By knowing the direction of the beach, the students could try to work out where the debris was coming from.

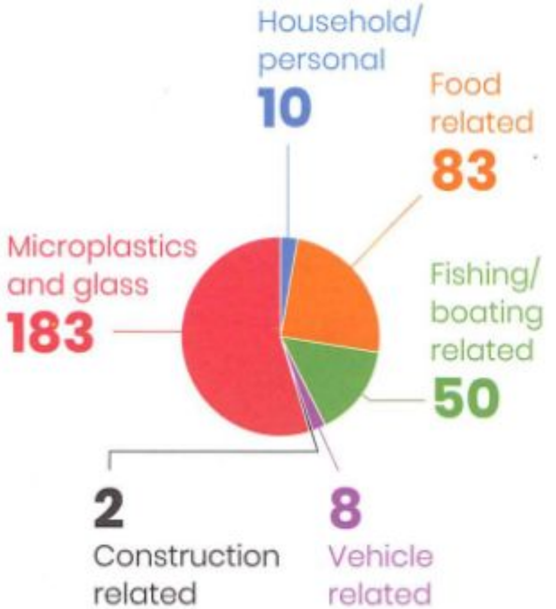
The students and adults worked together in small groups. They were given gloves and bags to collect the debris. Once they had finished, they sorted the debris into the following categories:

- **Microplastics and glass**
- **Household/personal:** clothing, furniture, cigarette butts, etc.
- **Food related:** straws, bottles, lollipop sticks, etc.
- **Fishing/boating related:** rope, fishing nets, **mussel lanyards**, boat parts, etc.
- **Vehicle related:** parking tickets, tyres, car parts, etc.
- **Construction related:** building materials, etc.

CHECKING THE RESULTS

The results from the three schools showed some interesting differences. This suggested that the location of the beach affected the type of marine debris found.

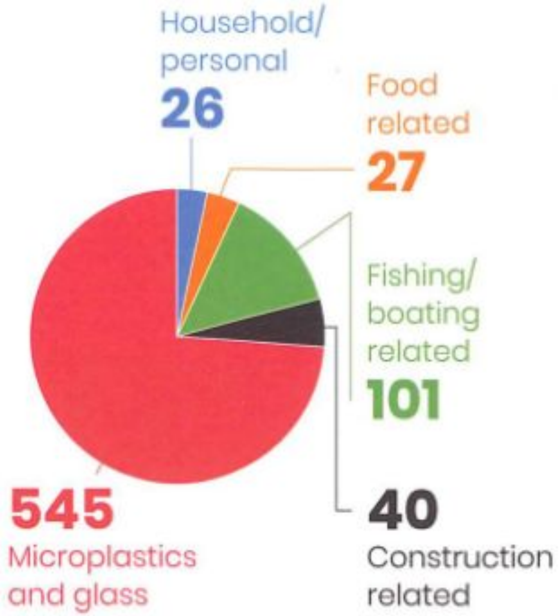
Mulberry Grove TOTAL ITEMS **336**



Mulberry Grove found lots of microplastics, glass, and food packaging.

The bay that Mulberry Grove cleaned faces Auckland. They found household debris from both island and off-island sources. They also found parking tickets that had floated all the way from Auckland!

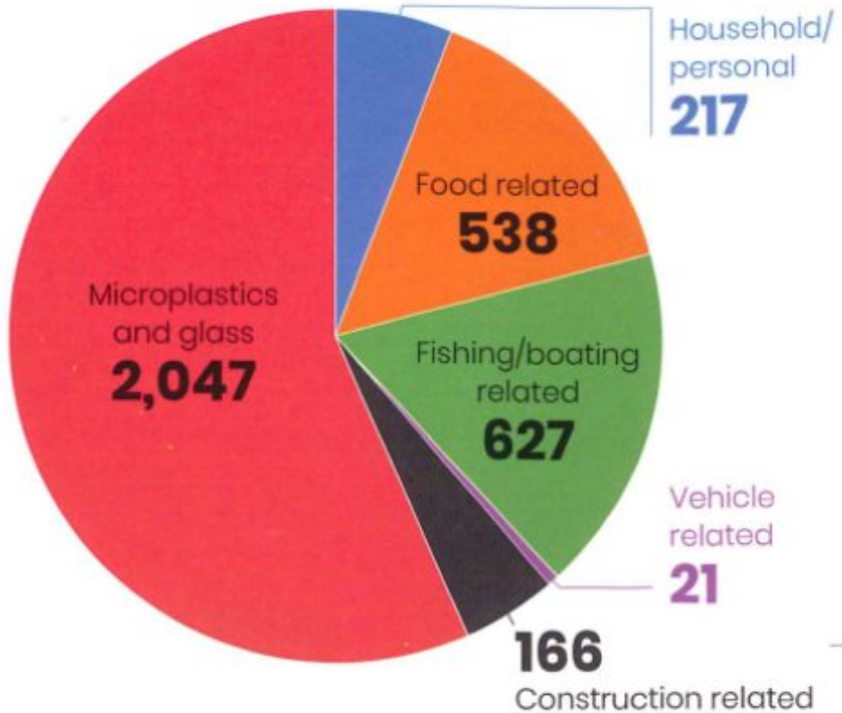
Kaitoke TOTAL ITEMS **739**



Kaitoke's data showed lots of microplastics. They also found a couch and a fridge!

Kaitoke Beach faces the Pacific Ocean, so these microplastics had likely travelled a long distance. Where do you think the couch and fridge came from?

Te Kura o Okiwi TOTAL ITEMS **3,616**



Te Kura o Okiwi's data showed that mussel lanyards are a big problem. They also found lots of rope, glass, microplastics, and even jandals.

Te Kura o Okiwi cleaned a variety of beaches that face different directions. The mussel lanyards were found near the island's mussel farms. The glass, rope, and jandals probably came from visiting boats.

SENDING A MESSAGE

The students chose creative and fun ways to share what they had found. They created art, sculptures, and wall displays from the marine debris. They told their stories in poems, songs, and even a play. They also made three videos.

The students shared their creations at a hui. Over one hundred people came, helping the students to spread their message about marine debris far and wide.

Headdress made from rope and mussel lanyards

Shark sculpture made from debris

FROM COMMUNICATION TO ACTION

The hui was just the beginning.
The students also:

ran community beach clean-ups

presented a talk about the project to over one hundred scientists at a science conference

worked with the mussel farm owners to find ways to stop lanyards from ending up on the beaches

made a presentation to over three hundred university students about how marine debris affects Aotea Great Barrier Island.

The students on Aotea Great Barrier Island are working hard to change their environment for the better. If they can do it, so can you!

GLOSSARY

environment – the surroundings in which an animal or plant lives

marine – found in the sea

microplastics – very small pieces of plastic, often formed when larger pieces of plastic break down

mussel lanyards – ropes used by fishermen for growing mussels

nutrients – substances in food that our bodies need in order to function

organisms – living things that can function on their own

toxic chemicals – substances that can poison or harm you